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## Introduction

The purpose of the Oscar Awards Database is to model and manage data related specifically to the Academy Awards (Oscars), one of the most prominent film festivals worldwide. Only information about the Oscar Awards, such as the annual ceremony, award categories, nominated films, nominees, and their nations, is stored and managed in the database.

The main objective of this database is to effectively arrange Oscar-related data for reporting and analysis. It makes it easier to spot trends like category winners, recurrent nominees, or films that receive several nominations. Python was incorporated into the system for database connectivity and data retrieval, and MySQL (Ubuntu) was used for implementation.

## Entity Sets and why use them?

After analysing how the Oscar Awards function, the entities and relationships in the database were created. Relationships explain the rational connections between the entities, while each entity symbolises a fundamental element of the ceremony. Since only the Oscar Awards are modelled by the system, all entities and links directly reflect information related to that event.

The Entities:

- Ceremony- this contains details of the Oscar Awards event such as the ceremony year, date and venue of each year.
- Category- this contains details such as the category name, description, and award Type.
- Nomination- this serves as a link between the other organisations, stating which nominee, movie, and category are associated with a specific ceremony.
- Film- this contains details of each film that is nominated for the win such as the title, and year.
- Nominee- this represents the people (or organisations) who have received an official Oscar nomination.
- Country- the nationality of each nominee is noted in order to emphasise the Oscars' global representation.

## Entity Tables

| Entity            | Primary Key   | Other Attributes                                                                   |
|-------------------|---------------|------------------------------------------------------------------------------------|
| <b>Ceremony</b>   | Ceremony_Id   | Ceremony_Year,<br>Ceremony_Date,<br>Ceremony_Venue                                 |
| <b>Category</b>   | Category_Id   | Category_Name,<br>Category_AwardType                                               |
| <b>Nomination</b> | Nomination_Id | Nomination_Role,<br>Nomination_Nominees,<br>Nomination_Title,<br>Nomination_Result |
| <b>Film</b>       | Film_Id       | Film_Title,<br>Film_Year                                                           |
| <b>Nominee</b>    | Nominee_Id    | Nominee_FirstName,<br>Nominee_LastName,<br>Nominee_Birthdate                       |
| <b>Country</b>    | Country_Name  | Country_Nationality                                                                |

---

To accurately capture all relevant details regarding the Oscars, nominations, and ceremonies, each entity was selected to represent a significant aspect of the event.

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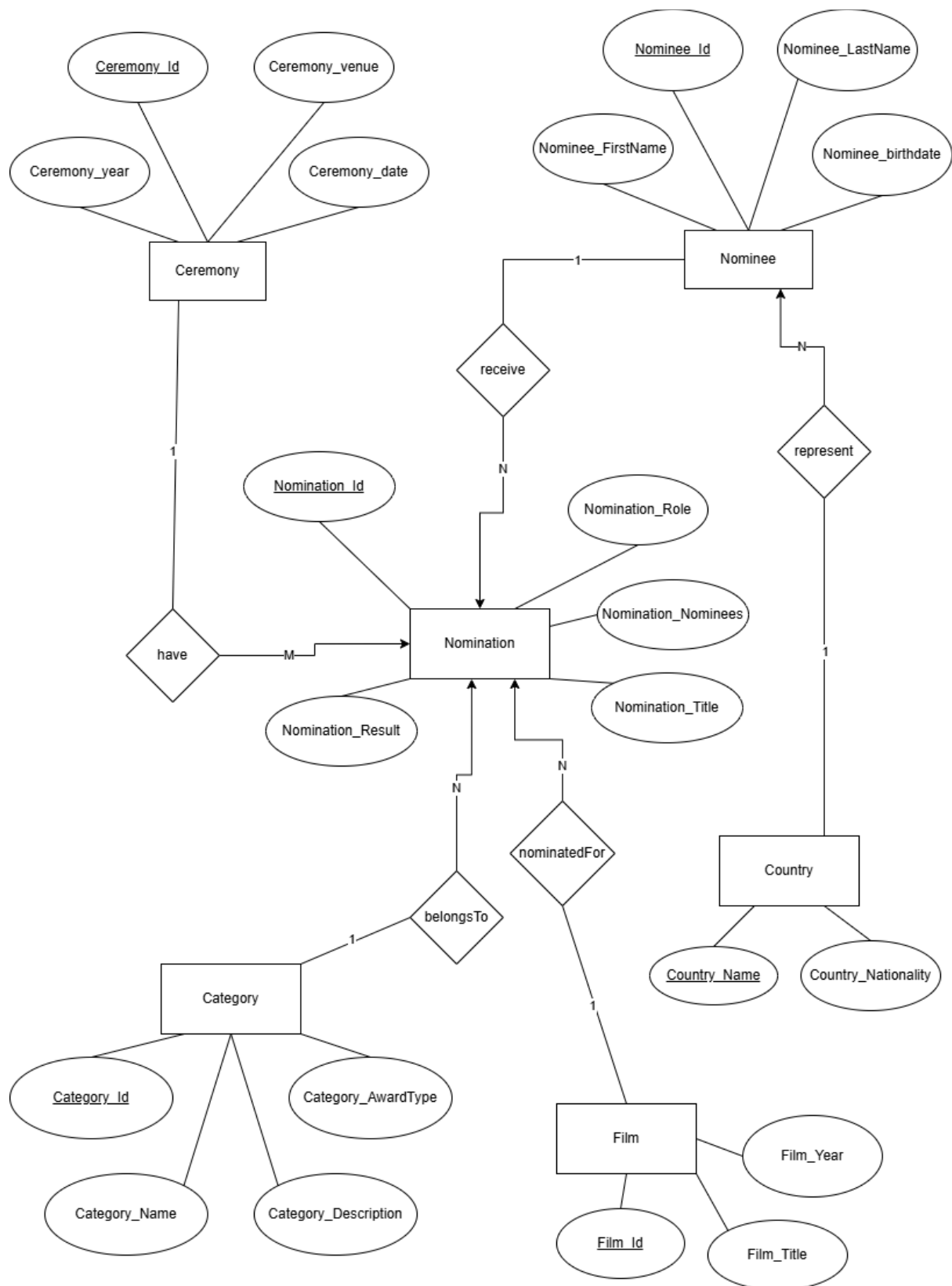
## Relationship Sets

| Relationship Sets   | Between                | Attributes                                                      |
|---------------------|------------------------|-----------------------------------------------------------------|
| <b>have</b>         | Ceremony -> Nomination | Ceremony_Id (FK),<br>Nomination_Id (FK)                         |
| <b>belongsTo</b>    | Category -> Nomination | Category_Id (FK),<br>Nomination_Id(FK)                          |
| <b>nominatedFor</b> | Film -> Nomination     | Film_Id (FK),<br>Nomination_Id (FK)                             |
| <b>receive</b>      | Nominee -> Nomination  | Nominee_Id (FK),<br>Nomination_Id (FK),<br>Nomination_Role (FK) |
| <b>represent</b>    | Country -> Nominee     | Country_Name (FK),<br>Nominee_Id (FK)                           |

## Cardinality Constraints & Participation Constraints

| Relationship Set    | Cardinality Constraints                                                                        | Participation Constraints                                                                                 |
|---------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>have</b>         | 1:N -> one ceremony can have many nominations, but many nominations are had in one ceremony.   | A ceremony cannot exist without nominations. Every nomination must belong to a ceremony.                  |
| <b>belongsTo</b>    | 1:N -> one category can have many nominations that belongs to it.                              | A category cannot exist without a nomination. A nomination cannot exist without a category.               |
| <b>nominatedFor</b> | 1:N -> one film can be nominated for many nominations, and many nominations can have one film. | A film can exist without being nominated, but a nomination cannot exist without a film.                   |
| <b>receive</b>      | 1:N -> one nominee can receive many nominations.                                               | A nominee cannot exist even without current nominations, and a nomination cannot exist without a nominee. |
| <b>represent</b>    | 1:N -> one country can represent many nominees, but each nominee belongs to one country.       | A country can exist without nominee, but nominee cannot exist without a country.                          |

## Entity Relationship Diagram



The Oscar Awards Database's structure is shown in the ER diagram above using Chen's notation. Relationships are represented by diamonds, attributes by ovals, and each entity is represented by a rectangle. The diagram shows how the nominee, ceremony, category, and movie entities are all connected. Cardinalities (1:N) ensure effective data organisation throughout the Oscar Awards system by demonstrating that a single record from the main entity can relate to numerous records in the smaller entity.

## Implementation

### Show databases

```
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| OscarAwards |
| company |
| dswork |
| dsworks |
| information_schema |
| just |
| mysql |
| performance_schema |
| sys |
+-----+
9 rows in set (0.01 sec)

mysql>
```

## Show tables

```
mysql> SHOW TABLES;
+-----+
| Tables_in_OscarAwards |
+-----+
| Category               |
| Ceremony               |
| Country                |
| Film                   |
| Nomination             |
| Nominee                 |
+-----+
6 rows in set (0.00 sec)

mysql>
```

## Category Table

```
mysql> DESC Category;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Category_Id    | int           | NO   | PRI | NULL    |       |
| Category_Name  | varchar(85)   | NO   |     | NULL    |       |
| Category_AwardType | varchar(85)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

## Ceremony Table

```
mysql> DESC Ceremony;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Ceremony_Id    | int           | NO   | PRI | NULL    |       |
| Ceremony_Year  | int           | NO   |     | NULL    |       |
| Ceremony_Date  | date          | YES  |     | NULL    |       |
| Ceremony_Venue | varchar(85)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)

mysql>
```

## Country Table

```
mysql> DESC Country;
```

| Field               | Type         | Null | Key | Default | Extra |
|---------------------|--------------|------|-----|---------|-------|
| Country_Name        | varchar(100) | NO   | PRI | NULL    |       |
| Country_Nationality | varchar(100) | YES  |     | NULL    |       |

2 rows in set (0.00 sec)

```
mysql>
```

## Film Table

```
mysql> DESC Film;
```

| Field      | Type         | Null | Key | Default | Extra |
|------------|--------------|------|-----|---------|-------|
| Film_Id    | int          | NO   | PRI | NULL    |       |
| Film_Title | varchar(150) | NO   |     | NULL    |       |
| Film_Year  | int          | YES  |     | NULL    |       |

3 rows in set (0.00 sec)

```
mysql>
```

## Nomination Table

```
mysql> DESC Nomination;
```

| Field               | Type         | Null | Key | Default | Extra |
|---------------------|--------------|------|-----|---------|-------|
| Nomination_Id       | int          | NO   | PRI | NULL    |       |
| Nomination_Role     | varchar(100) | YES  |     | NULL    |       |
| Nomination_Nominees | varchar(200) | YES  |     | NULL    |       |
| Nomination_Title    | varchar(100) | YES  |     | NULL    |       |
| Nomination_Result   | varchar(100) | YES  |     | NULL    |       |

5 rows in set (0.01 sec)

```
mysql>
```

## Nominee Table

```
mysql> DESC Nominee;
```

| Field             | Type         | Null | Key | Default | Extra |
|-------------------|--------------|------|-----|---------|-------|
| Nominee_Id        | int          | NO   | PRI | NULL    |       |
| Nominee_FirstName | varchar(100) | NO   |     | NULL    |       |
| Nominee_LastName  | varchar(100) | YES  |     | NULL    |       |
| Nominee_Birthdate | date         | YES  |     | NULL    |       |

```
4 rows in set (0.00 sec)
```

```
mysql> █
```

## The sample data

MySQL was used to implement the database on Ubuntu. SQL DDL commands (CREATE TABLE, ALTER TABLE, and etc.) were used to create all tables and constraints, and 'LOAD DATA LOCAL INFILE' was used to import CSV files.

### Count how many rows

```
mysql> SELECT COUNT(*) AS totalRowsCeremony FROM Ceremony;
+-----+
| totalRowsCeremony |
+-----+
|                150 |
+-----+
1 row in set (0.01 sec)

mysql> SELECT COUNT(*) AS totalRowsCategory FROM Category;
+-----+
| totalRowsCategory |
+-----+
|                150 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT COUNT(*) AS totalRowsCountry FROM Country;
+-----+
| totalRowsCountry |
+-----+
|                150 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT COUNT(*) AS totalRowsFilm FROM Film;
+-----+
| totalRowsFilm |
+-----+
|                200 |
+-----+
```

```
mysql> SELECT COUNT(*) AS totalRowsNominee FROM Nominee;
+-----+
| totalRowsNominee |
+-----+
|                250 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT COUNT(*) AS totalRowsNomination FROM Nomination;
+-----+
| totalRowsNomination |
+-----+
|                 300 |
+-----+
1 row in set (0.00 sec)

mysql> 
```

## Preview Actual Data

```
mysql> SELECT * FROM Film LIMIT 5;
+-----+-----+-----+
| Film_Id | Film_Title | Film_Year |
+-----+-----+-----+
| 1 | Moon from Paper Moon | 1982 |
| 2 | Empire in Last River | 1951 |
| 3 | Fields of the Eternal Moon | 1980 |
| 4 | Eternal Dreams | 1982 |
| 5 | Horizon of Feral Summer | 1950 |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Nomination LIMIT 5;
+-----+-----+-----+-----+-----+
| Nomination_Id | Nomination_Role | Nomination_Nominees | Nomination_Title | Nomination_Result |
+-----+-----+-----+-----+-----+
| 1 | Writer | Zoe Saldana | Moon of Feral Dreams | Leonardo DiCaprio |
| 2 | Composer | Taika Waititi | Storm and the Hidden Waves | Mahesh Babu |
| 3 | Editor | Michael Fassbender | Rising Voices | Daniel Craig |
| 4 | Composer | Russell Crowe | Fields of the Eternal Moon | Russell Crowe |
| 5 | Producer | Shonda Rhimes | Winter of Atomic Empire | Shonda Rhimes |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> 
```

## Nominations By The Year

```
mysql> SELECT
->   f.Film_Year AS Year,
->   n.Nomination_Title AS FilmTitle,
->   n.Nomination_Nominees AS NomineeName,
->   n.Nomination_Role AS Role,
->   n.Nomination_Result AS Result
-> FROM Nomination n
-> JOIN Film f
->   ON n.Nomination_Title = f.Film_Title
-> ORDER BY f.Film_Year DESC, n.Nomination_Title;
```

| Year | FilmTitle                   | NomineeName       | Role            | Result            |
|------|-----------------------------|-------------------|-----------------|-------------------|
| 2023 | Fields and the Silver Waves | Choi Woo-shik     | Producer        | Choi Woo-shik     |
| 2022 | Sun from Crimson Empire     | Tobey Maguire     | VFX Supervisor  | Catherine Deneuve |
| 2022 | Sun from Crimson Empire     | Catherine Deneuve | Composer        | Catherine Deneuve |
| 2022 | Waves in Hidden Sun         | Nicole Kidman     | Editor          | Nicole Kidman     |
| 2022 | Waves in Hidden Sun         | Hugh Grant        | Cinematographer | Nicole Kidman     |
| 2021 | Dreams of Silver Empire     | Ranbir Kapoor     | Designer        | Ranbir Kapoor     |
| 2021 | Paths of the Crimson Shadow | Wagner Lamounier  | Writer          | Wagner Lamounier  |
| 2021 | Paths of the Crimson Shadow | Ben Whishaw       | Director        | Wagner Lamounier  |
| 2020 | Winter of Paper Horizon     | Tang Wei          | Editor          | Lupita Nyong'o    |
| 2020 | Winter of Paper Horizon     | Paul Rudd         | Composer        | Lupita Nyong'o    |
| 2020 | Winter of Paper Horizon     | Lupita Nyong'o    | Producer        | Lupita Nyong'o    |
| 2019 | City in Fallen Winter       | Cobie Smulders    | Cinematographer | Cobie Smulders    |
| 2019 | City in Fallen Winter       | Monica Bellucci   | Writer          | Cobie Smulders    |
| 2019 | City in Fallen Winter       | Karen Gillan      | Writer          | Cobie Smulders    |
| 2019 | House in First Horizon      | Nicole Kidman     | Actor           | Nicole Kidman     |
| 2019 | Neon Waves                  | Halle Berry       | Producer        | Halle Berry       |
| 2019 | River and the Eternal Moon  | Richard Madden    | VFX Supervisor  | Richard Madden    |
| 2019 | River and the Eternal Moon  | Yash Yash         | Director        | Richard Madden    |
| 2019 | River and the Eternal Moon  | Gong Yoo          | Composer        | Richard Madden    |

To group nominations by the year

```
mysql> SELECT
->     f.Film_Year AS Year,
->     COUNT(n.Nomination_Id) AS TotalNominations
-> FROM Nomination n
-> JOIN Film f ON n.Nomination_Title = f.Film_Title
-> GROUP BY f.Film_Year
-> ORDER BY f.Film_Year DESC;
```

| Year | TotalNominations |
|------|------------------|
| 2023 | 1                |
| 2022 | 4                |
| 2021 | 3                |
| 2020 | 3                |
| 2019 | 10               |
| 2018 | 5                |
| 2017 | 3                |
| 2016 | 4                |
| 2015 | 3                |
| 2014 | 1                |
| 2012 | 2                |
| 2010 | 4                |
| 2005 | 5                |
| 2004 | 7                |
| 2003 | 8                |
| 2001 | 5                |
| 1999 | 3                |
| 1998 | 2                |
| 1997 | 5                |
| 1996 | 7                |
| 1994 | 4                |
| 1993 | 8                |
| 1992 | 2                |

## Advanced Features

### STORED PROCEDURES

Retrieves all the winners and their roles for a given ceremony year

```
mysql> CALL WinnersByYear(2016);
+-----+-----+-----+-----+
| Film_Title          | Film_Year | Role          | WinnerName    |
+-----+-----+-----+-----+
| Electric House      | 2016     | Composer      | Andrew Scott  |
| Paths of Crimson Waves | 2016     | VFX Supervisor | Mahesh Babu   |
| Silver Summer       | 2016     | Designer      | Ang Lee       |
+-----+-----+-----+-----+
3 rows in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> █
```

### Views

To support reporting, a view can be created

```
mysql> CREATE VIEW vw_winners AS
-> SELECT f.Film_Title, n.Nomination_Result AS WinnerName, f.Film_Year
-> FROM Film f
-> JOIN Nomination n ON f.Film_Title = n.Nomination_Title
-> WHERE n.Nomination_Result = n.Nomination_Nominees;
Query OK, 0 rows affected (0.04 sec)

mysql> █
```

The purpose of this is to simplify retrieving all winners from a single query

## Python Connectivity

Python 3 (Ubuntu) was used to connect to the MySQL database via the mysql-connector-python library

```
import mysql.connector

conn = mysql.connector.connect(
    host="127.0.0.1",
    user="root",
    password="YOURPASSWORD",
    database="OscarAwards"
)

cursor = conn.cursor()
cursor.execute("SELECT COUNT(*) FROM Film;")
print("Total Films:", cursor.fetchone()[0])

cursor.callproc("WinnersByYear", (2016,))
for result in cursor.stored_results():
    for row in result.fetchall():
        print(row)

cursor.close()
conn.close()
```

## Conclusion

The entire database system lifecycle, from conceptual ER design to implementation, population, and external connectivity, is illustrated by the OscarAwards Database project. By applying good normalisation principles, the schema successfully simulates intricate award relationships found in the real world.

The use of triggers, stored procedures, and Python connectivity demonstrates sophisticated programming and SQL abilities. The system demonstrates the ability to plan, execute, and maintain a relational database for a practical situation by successfully fulfilling all assignment requirements.